RITESH SETH

Curriculum Vitae

B-513, 5th Floor, R&D Building
IIIT Delhi, New Delhi, India
riteshs@iiitd.ac.in

Research Experience

Aug.'19–Dec.'21 **Research Project in Algorithm and Computational Geometry**, *Indian Statistical Institute, Kolkata (India)*, supervisor : Prof. Subhas C. Nandy, (Focused on : Minimum height watchtower on terrain, Shortest path with obstacle violation, 1-violation path in simple polygon)

Education

2022–Cont. **PhD. in Computer Science**, *Indraprastha Institute of Information Technology*, Delhi (India)

2017–2019 **M.Sc. in Computer Science**, *Ramakrishna Mission Vivekananda Educational and Research Institute*, Belur Math, Howrah, West Bengal (India)

2014–2017 **B.Sc.(hons.) in Computer Science**, *University of Calcutta*, Kolkata (India)

2012–2014 **Higher Secondary Examination (10+2)**, *West Bengal Council of Higher Secondary Education*, India

2010–2012 **Secondary Examination**, *West Bengal Board of Secondary Education*, India

Theses

masters **A survey on homomorphisms of signed graphs**, *supervisors: Prof. Sagnik Sen and Prof. Soumen Nandi*

Publications

conference **"Near Optimal Online Bipartite Matching on the Line with Amortized Logarithmic Recourse"**, *Syamantak Das, Sharath Raghvendra and Ritesh Seth*, FSTTCS, 2026

conference **"Efficient algorithms for Incremental Metric Bipartite Matching"**, *Ritesh Seth, Mrinal Garg, Sujoy Bhore, Sharath Raghvendra and Syamantak Das*, ICLR, 2026

journal **"Acrophobic guard watchtower problem"**, *Ritesh Seth, Anil Maheshwari and Subhas C Nandy*, Computational Geometry, 2023

conference **"An Acrophobic Guard Watchtower Problem on Terrains."**, *Ritesh Seth, Anil Maheshwari and Subhas C Nandy*, CCCG, 2021

conference **"The Relative Signed Clique Number of Planar Graphs is 8."**, *Sandip Das, Soumen Nandi, Sagnik Sen and Ritesh Seth.*, CALDAM, 2019

Teaching Experience

Aug.'25–Dec.'25 **Working as a TA in Usable Security and Privacy (CSE547)**, Department of Computer Science and Engineering
Indraprastha Institute of Information Technology Delhi (IIITD)

Jan.'25–May.'25 **Worked as a TA in Analysis and Design of Algorithms (CSE222)**, Department of Computer Science and Engineering
Indraprastha Institute of Information Technology Delhi (IIITD)

Aug.'24–Dec.'24 **Worked as the TA in Modern Algorithm Design (CSE519)**, Department of Computer Science and Engineering
Indraprastha Institute of Information Technology Delhi (IIITD)

Jan.'24–May.'24 **Worked as a TA in Analysis and Design of Algorithms (CSE222)**, Department of Computer Science and Engineering
Indraprastha Institute of Information Technology Delhi (IIITD)

Aug.'23–Dec.'23 **Worked as the TA in Modern Algorithm Design (CSE519)**, Department of Computer Science and Engineering
Indraprastha Institute of Information Technology Delhi (IIITD)

Jan.'23–May.'23 **Worked as a TA in Analysis and Design of Algorithms (CSE222)**, Department of Computer Science and Engineering
Indraprastha Institute of Information Technology Delhi (IIITD)

Aug.'22–Dec.'22 **Worked as a TA in Introduction to Graduate Algorithms (CSE525)**, Department of Computer Science and Engineering
Indraprastha Institute of Information Technology Delhi (IIITD)

Jan.'22–May.'22 **Worked as a TA in Analysis and Design of Algorithms (CSE222)**, Department of Computer Science and Engineering
Indraprastha Institute of Information Technology Delhi (IIITD)

Jan.'20–May.'20 **Information Technology in Sports (MSSCT-206)**, *(Basics of Python, basics of ML using Python, LaTeX)*, M.Sc sports 19-21 batch, Department of Sports Science and Yoga
Ramakrishna Mission Vivekananda Educational and Research Institute (RKMVERI)

Jul.'19–Nov.'19 **Computer Application (PGDY-CP-108)**, *(Basics of Computer Applications including Word, Excel and PowerPoint)*, PGDY 2019-20 batch, Department of Sports Science and Yoga
Ramakrishna Mission Vivekananda Educational and Research Institute (RKMVERI)

Projects and Internships

Aug.'19-Dec.'21. **Research Project**, *Indian Statistical Institute, Kolkata (India)*, supervisor : Prof. Subhas C. Nandy

Feb.-Mar.'19 **Research Internship**, *Indian Institute of Technology, Dharwad (India)*, supervisor : Prof. Sagnik Sen

Dec.'18-Jan.'19 **Research Internship**, *Indian Statistical Institute, Kolkata (India)*, supervisor : Prof. Soumen Nandi

Sep.'18 **Presentation in open problem session in "ELGGA"**, *BITS Pilani, Hyderabad (India) in association with ACSDM (Association for Computer Science and Discrete Mathematics)*

June.'18 **Summer Project**, *Ramakrishna Mission Vivekananda Educational and Research Institute, Howrah (India)*, supervisor : Prof. Sagnik Sen

Funding / Scholarships

June.'19 **Obtained funding for being TA at ACM summer school on "Geometric Algorithms and Their Applications"**, *NISER, Bhubaneswar (India)*

Feb.-Mar.'19 **Obtained funding for Research Internship**, *Indian Institute of Technology, Dharwad (India)*, supervisor : Prof. Sagnik Sen

Sep.'18 **Was selected and obtained funding for "ELGGA"**, *BITS Pilani, Hyderabad (India) in association with ACSDM (Association for Computer Science and Discrete Mathematics)*

2017-19 **Swami Vivekananda Merit-Cum-Means Scholarship**, *Government of West Bengal*, India

Technical Skills

Programming Languages

C, C++, Python, CUDA, Visual Basic, Visual C++

Tools

Scikit-Learn, Keras, PyTorch

Workshop / Conference Attended

April.'26 **The Fourteenth International Conference on Learning Representations (ICLR)**, *Riocentro Convention and Event Center, Rio de Janeiro (Brazil)*, Presenter (Efficient algorithms for Incremental Metric Bipartite Matching)

Dec.'24 **Frontiers of Geometric Algorithms**, *IISc, Bengaluru (India)*

Oct.'24 **An ACM IndiCS Seminar on "Continuous Methods in Discrete Optimization and Complexity"**, *Infosys Mysore campus, Karnataka (India)*

July.'23 **Recent Trends in Algorithms**, *NISER, Bhubaneswar (India)*

Dec.'22 **Foundations of Software Technology and Theoretical Computer Science (FSTTCS) on Algorithms under Uncertainty**, *IIT, Madras (India)*

Aug.'21 **33rd Canadian Conference on Computational Geometry (CCCG)**, *Halifax, Nova Scotia (Canada)*, Speaker (An Acrophobic Guard Watchtower Problem on Terrains)

June.'19 **ACM summer school on "Geometric Algorithms and Their Applications"**, *NISER, Bhubaneswar (India)*, Organizing Committee-Volunteer

Jan.'19 **Series of lectures: AGraHo Project**, *Organized by Ramakrishna Mission Vivekananda Educational and Research Institute, Belur Math, Howrah (India)*

Feb.'18 **Workshop on Big Data Analytics**, *Ramakrishna Mission Vivekananda Educational and Research Institute, Belur Math, Howrah (India)*, Organizing Committee-Volunteer